

Bishal Neupane

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RESEARCH INTERESTS

Precision agriculture, remote sensing and satellite imagery analysis, geospatial data science, crop monitoring systems, plant growth regulation, AI/ML applications in agricultural systems, biosystems engineering, and data-driven approaches to sustainable food production in South Asia.

EDUCATION

- Agriculture and Forestry University (AFU)** Rampur, Nepal
Bachelor of Science in Agriculture; GPA: 3.12 *April 2021 – Nov 2025*
 - Thesis:** Investigated effects of Auxin (NAA) and Gibberellic Acid (GA₃) on okra growth and yield under RCBD experimental design; GA₃ at 100 ppm identified as the most effective treatment.

RESEARCH EXPERIENCE

- PMAMP Morang (Prime Minister Agriculture Modernization Project)** Nepal
Agricultural Research Intern *Jan 2025 – Jul 2025*
 - Experimental Design & Field Research:** Designed and executed RCBD field experiments on 50+ okra plants across 3 hormone concentration levels; collected 200+ morphological and yield data points for statistical analysis.
 - Statistical Analysis:** Applied ANOVA, LSD, and treatment comparison methods in R; achieved 15% yield improvement through data-driven hormone treatment optimization; produced publication-ready visualizations.
- Open Mapping Hub Asia Pacific** Remote
Open Mapping Guru – GIS & Geospatial Research *Jan 2024 – Present*
 - Geospatial Data Analysis:** Validated 5,000+ geographic data points achieving 20% improvement in mapping accuracy; corrected 300+ data points across 15+ regional datasets using Python-based GIS tools.
 - International Collaboration:** Coordinated with 8 international researchers, increasing regional mapping data contributions by 40% over 6 months.
- Independent Research & Development** Remote
Precision Agriculture & AI Researcher *Aug 2024 – Present*
 - Remote Sensing Research:** Conducted multi-temporal satellite analysis using Google Earth Engine and Sentinel-2 imagery across 500+ hectares; computed NDVI, NDWI, SAVI, GCI, and PSRI indices for crop health and environmental monitoring.
 - AI/ML in Agriculture:** Developed CNN-based crop disease detection system using TensorFlow and MobileNetV2; contributed GPS-tagged field data to Nepal Agricultural Disease Dataset (NADD).

RESEARCH & TECHNICAL PROJECTS

- Agricultural Statistical Analysis Framework:** Engineered R-based statistical pipeline for multi-variable RCBD agricultural experiments using tidyverse, ggplot2, and agricolae. Implemented automated ANOVA, Duncan's MRT, and 4-panel diagnostic visualizations; achieved $p < 0.001$ significance for 96% of variables across 30+ collaborative research datasets.
- Precision Agriculture Analysis – Kaski, Nepal:** Applied Google Earth Engine with Sentinel-2 imagery to classify 500+ hectares of agricultural land using K-Means clustering (85% accuracy across 6 land-cover classes); computed 5 vegetation indices for 200+ farm plots supporting precision farming and crop monitoring decisions.
- NDWI Environmental Analysis – Pokhara Lekhnath:** Conducted multi-temporal NDWI satellite analysis using Sentinel-2 on Google Earth Engine to track seasonal water presence, identify persistent negative NDWI patterns, and map environmental changes supporting urban resilience and conservation research.
- Crop Disease Detection System:** Developed mobile AI system using Flutter, FastAPI, TensorFlow, and MobileNetV2 CNN for field-based crop disease diagnosis. Integrated FAO/USDA treatment citations, offline TFLite inference, and GPS-tagged data contributing to Nepal Agricultural Disease Dataset (NADD).

TECHNICAL SKILLS

- **Research Computing:** R (tidyverse, ggplot2, agricolae, gvlma, R Markdown), Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), ANOVA, RCBD, Duncan's MRT, PCA, Time-Series Analysis, Regression
- **Crop Modelling:** DSSAT v4.8.5 (CROPGRO, CERES), crop simulation, genetic coefficient calibration, decision support systems for agrotechnology transfer
- **Remote Sensing & GIS:** Google Earth Engine, Sentinel-2, Landsat, MODIS, NDVI/NDWI/SAVI/GCI/PSRI, K-Means Classification, GIS Analysis, OpenStreetMap, QGIS
- **AI & Machine Learning:** TensorFlow, Keras, MobileNetV2 (CNN), NLP, RAG Pipelines (ChromaDB), Google Gemini AI, ML Forecasting, Pattern Recognition
- **Data Engineering & Dev:** FastAPI, Django, Flask, React, Node.js, Flutter, PostgreSQL, MongoDB, SQLite, Docker, Git, GitHub
- **Scientific Tools:** Jupyter Notebook, RStudio, Anaconda, Google AI Studio, Google Cloud Platform, Power BI, Tableau, Advanced Excel

PUBLICATIONS & MANUSCRIPTS

Accepted / In Press

- Neupane, B. et al. (2026). Rice Mechanization in Nepal: Status, Challenges, and Pathways Toward Sustainable Agricultural Transformation—A Review. *Tropical Agrobiodiversity (TRAB)*, Zibeline International Publishing. (Accepted May 2026, in press)
- Neupane, B. et al. (2026). Agricultural Mechanization Status and Adoption Constraints in Maize Farming: Evidence from a Hill Municipality of Nepal. *Tropical Agrobiodiversity (TRAB)*, Zibeline International Publishing. (Accepted May 2026, in press)

Under Review

- Neupane, B. et al. Pesticide Retailers as De Facto Extension Agents: Knowledge, Practices, and Implications for IPM Adoption among Vegetable Growers in Bhairahawa, Nepal. *Journal of Agriculture and Rural Development in the Tropics and Subtropics (JARTS)*. (Revised manuscript under review)
- Neupane, B., Thakur, L., Katuwal, L., & Chaudhary, M. Effect of Organic and Inorganic Fertilizer Amendments on Growth and Yield of Spring Rice (*Oryza sativa* L.) in Eastern Terai, Nepal. *Turkish Journal of Agriculture–Food Science and Technology (TURJAF)*. (Major revisions submitted June 2026)

CERTIFICATIONS

- **Basics of Crop Modeling: Fundamentals and Applications:** NAXA & YES Agriculture – 5-day online training, Feb 10–14, 2025
- **Associate Data Scientist with Python:** DataCamp – Advanced Python data science libraries, statistical modeling, and ML workflows
- **Google Data Analytics Certificate:** Google – Data analysis, SQL, R programming, and visualization techniques
- **End-to-End Google Earth Engine:** Geospatial datasets, dynamic mapping, watershed and land-cover classification for Nepal
- **Elements of AI:** MinnaLearn – AI fundamentals and machine learning concepts

LANGUAGES

- **English:** Professional academic proficiency – research writing, scientific communication, and international collaboration
- **Nepali:** Native speaker

INVOLVEMENT

- **IAAS Nepal LC Puranchaur** Puranchaur, Kaski
VP of Communication *Feb 2022 – Nov 2025*
 - **Leadership:** Managed organizational communications, social media strategy, meetings, and data reporting for IAAS Nepal LC Puranchaur.
- **GAVS Puranchaur** Puranchaur, Kaski
Vice-Coordinator *Nov 2021 – Nov 2025*
 - **Field Coordination:** Organized workshops, managed social media content, and served as field coordinator for the agricultural student organization.

ACHIEVEMENTS

- **National Exchange Coordinator Meeting:** Attended the “National Exchange Coordinator Meeting 2022” organized by IAAS Nepal LC Puranchaur.